Sports Tournament Ranking System

1. Problem Statement

Develop a Python program to store match results, calculate team points, apply tie-break rules, and display tournament rankings.

2. Objectives

- Store match results.
- Calculate team points.
- Calculate goal difference.
- Apply tie-break rules.
- Display team rankings.

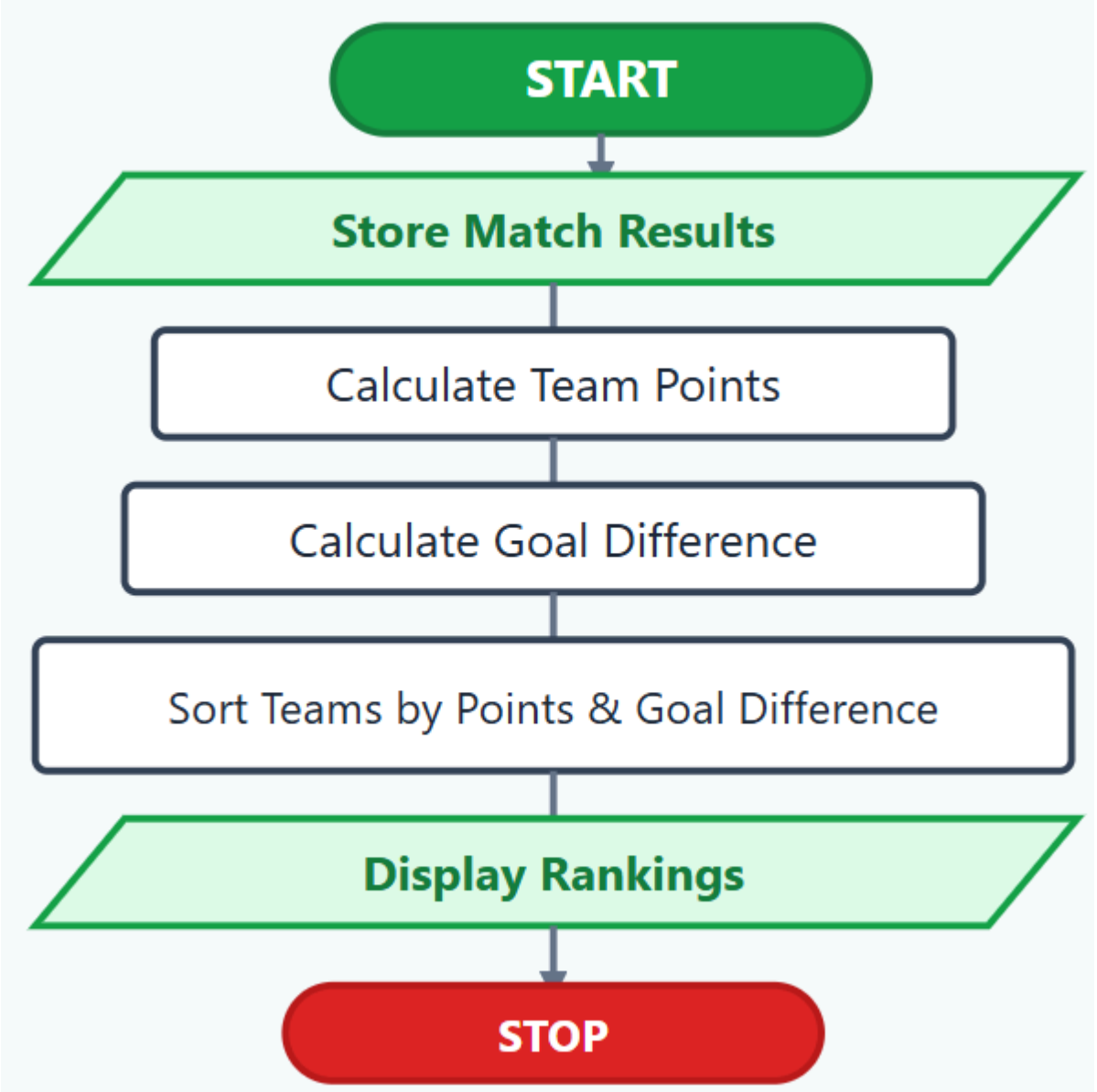
3. Assumptions

- Win = 3 points.
- Draw = 1 point.
- Loss = 0 points.
- Goal difference is used as the first tie-breaker.
- Higher points and goal difference get higher rankings.

4. Algorithm

1. Start.
2. Store match results.
3. Initialize team points and goal difference.
4. Calculate points for each match.
5. Update goal difference.
6. Sort teams by points and goal difference.
7. Display rankings.
8. Stop.

5. Flowchart



None

6. Pseudocode

None

```
START

Store match results

FOR each match:
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    IF team 1 wins:
        Add 3 points to team 1
    ELSE IF match is draw:
        Add 1 point to both teams
    ELSE:
        Add 3 points to team 2

    Calculate goal difference

Sort teams by:
    Points descending
    Goal difference descending

Display rankings

STOP
```

8. Sample Output

```
None
TOURNAMENT RANKINGS

1 India Points: 4 Goal Difference: 2
2 Australia Points: 3 Goal Difference: 0
3 England Points: 1 Goal Difference: -2
```

9. Test Cases

Test	Match Result	Expected Result
1	3–1	Winner gets 3 points
2	2–2	Both get 1 point

3	0–2	Second team gets 3 points
4	Equal points	Compare goal difference
5	Different points	Higher points ranked first

10. Conclusion

The Sports Tournament Ranking System calculates team points and goal differences, applies tie-break rules, and displays rankings using Python lists, dictionaries, loops, and sorting.